

Nitin Yadav

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Summary

Applied AI engineer and data scientist with expertise in deep learning, computer vision, GenAI/LLMs, statistics, and causal inference. Proven record of building end-to-end AI/ML systems, translating research into scalable, real-world solutions.

Education

- Auburn University** Auburn, AL, USA
Master of Science in Artificial Intelligence Engineering Jun 2025 – May 2026 (Expected)
Master of Science in Probability and Statistics Aug 2022 – Dec 2023
- BITS Pilani** Hyderabad, India
Bachelor of Engineering in Chemical Engineering Jul 2016 – May 2020

Technical Skills

- Coursework:** Machine Learning, Deep Learning, GenAI & LLMs, Evolutionary Algorithms, Causal Inference, Statistical Modeling, Business Analytics, Multivariate Analysis, Linear Algebra, Calculus, Numerical Analysis
- Tools:** PyTorch, TensorFlow, Transformers, CUDA, CLIP, XGBoost, OpenCV, Pandas, Numpy, Python, R, SQL

Professional Experience

- JP Morgan Chase – Analytic Solutions Analyst**, Mumbai, India Jun 2024 – Aug 2024
 - Performed root-cause analysis, process engineering, and operational improvements for the Equity Ops team.
 - Enabled automatic reconciliation for SBL processes using Alteryx workflow, reducing workload equivalent to 1.5 FTE.
- Fractal Analytics – Data Analyst, Project Manager**, Mumbai, India Jun 2020 – Jul 2022
 - Built an ML pipeline (DAGs, scikit-learn) to predict P&G warehouse ETAs using historical data, improved accuracy by 15%.
 - Led the CLR project expansion by 20× and scaled a cross-functional team from 1 to 5 as SME/Scrum Manager, managed end-to-end client relations and strategy.
 - Developed Python-based algorithms and ETL pipelines to automate data analysis and reporting, formulated 4 key KPIs for monitoring compliance at inference and supporting leadership decision-making for customising products.

Research Experience

- Harbert College of Business, Auburn University – Research Assistant**, Auburn, AL (part-time) Sep 2022 – May 2025
 - Examined causal impact of GenAI adoption on Stack Overflow using network science and Difference-in-Differences, identifying about 60% change in user engagement.
 - Documented long-run effects on engagement, including a 2.5× decrease in clustering, 3× increase in node death rate.
 - Assisted in introducing the concept of Functional Form Misspecification in statistical modeling to address endogeneity.
- Innovation and Research Commons, Auburn University – Data Consultant**, Auburn, AL (part-time) Aug 2022 – Dec 2023
 - Automated extraction of course data from university-wide LMS (Canvas) to estimate dynamic weekly workloads.
 - Consulted 3-5 graduate students and faculty on data analysis, specializing in debugging R and Python scripts.

Academic Projects

- Autonomous Agent Learning with Evolutionary Algorithms** Aug 2025 – Dec 2025
 - Engineered a tree-based Genetic Programming framework to evolve autonomous controllers for the GPac game.
 - Developed a competitive co-evolutionary algorithm to evolve adversarial Pac-Man and Ghost agents simultaneously.
 - Designed agentic decision-making logic using initialization, mutation, and selection to deploy reinforcement learning.
 - Applied multi-objective optimization using Pareto-front analysis to balance score vs complexity via parsimony pressure.
- Zero-Shot Object Counting** Aug 2025 – Dec 2025
 - Implemented the VA-Count framework enabling class-agnostic object counting without human-labeled exemplars.
 - Automated exemplar mining using GroundingDINO and a CLIP-based classifier for high-quality single-object examples.
 - Achieved state-of-the-art performance on FSC147 dataset, monitored training via Weights & Biases, with data, code, and trained weights publicly available on GitHub for reproducibility.
- Object Recognition and Tracking** May 2025 – Aug 2025
 - Designed a real-time classical vision pipeline to detect(87% accuracy) and track fast-moving objects under occlusion.
 - Built and trained a CNN on MNIST for digit recognition and generalization to handwritten inputs with 99% accuracy.